

THE LOCAL WHITTAKER FACTOR AT A SPLIT LEVEL PRIME, AND AN INCOMPLETENESS IN THE STANDARD RECIPE FOR SCHOER'S

κ

ABSTRACT. Schofer's formula expresses averages of $\log\|\Psi_F\|$ over CM points on a Shimura curve in terms of quantities $\kappa_\eta(m)$ built from derivatives of Fourier coefficients of an incoherent Eisenstein series. The recipe used in practice to evaluate $\kappa_\eta(m)$ descends from Kudla–Rapoport–Yang, through Schofer and Errthum, to Yang; its central step enumerates the configurations that contribute to the derivative at the centre. We observe that this enumeration is derived under a restriction which Kudla–Rapoport–Yang state explicitly — that the Eisenstein series be built from sections given by the characteristic functions of the completions of $\mathcal{O}_k$ — and which is not satisfied by the lattices arising for an Eichler order of squarefree level N at a *split* level prime.

We compute the local Whittaker factor of the relevant N -scaled binary lattice at the level prime in closed form for every valuation, and show that its value at the critical point is $(N-1)\text{ord}_N(m)$. We then re-run the case analysis with this factor in place and find that it produces no additional contributing configuration, so that no $\log N$ can arise from $\kappa_\eta(m)$ for $m > 0$. Independently, a numerical evaluation of the vector-valued input form determines, exactly, a correction of the form $A_m \log N$ that the CM values demonstrably require. We record the resulting tension, the exact data any corrected formula must reproduce, and the routes we have eliminated.

1. INTRODUCTION

Let B be an indefinite quaternion algebra over $\mathbb{Q}$ of reduced discriminant D , let $\mathcal{O} \subset B$ be an Eichler order of squarefree level N with $(N, D) = 1$, and let $L = \{\alpha \in \mathcal{O} : \text{tr } \alpha = 0\}$ be the associated lattice of signature $(1, 2)$, so that $L^\vee/L \cong \mathbb{Z}/2 \times \mathbb{Z}/DN \times \mathbb{Z}/DN$ has order $2(DN)^2$ [2, Lemma 8]. For a weakly holomorphic vector-valued modular form F of weight $1/2$ and type ρ_L , Borchers' lift Ψ_F is a meromorphic modular form on $X_0^D(N)/W_{D,N}$, and Schofer's formula [5, 2] evaluates

$$\sum_{\tau \in \text{CM}(d)} \log|\psi_F(\tau)| = -\frac{|\text{CM}(d)|}{4} \sum_{\eta \in L^\vee/L} \sum_{m \geq 0} c_\eta(-m) \kappa_\eta(m),$$

valid under the hypotheses $O_{L,F}^+ = O_L^+$, $c_0(0) = 0$ and $c_\eta(m) \in \mathbb{Z}$ for $m \leq 0$.

The quantities $\kappa_\eta(m)$ are not given in [2]; the reader is referred to [1] and [6] for their evaluation. Both go back to the computation of Kudla–Rapoport–Yang [3], whose Proposition 2.8 identifies the configurations contributing to the derivative of an incoherent Eisenstein series at the centre and asserts that *these are the only cases which contribute*.

What this note contains. Section 2 traces that assertion to its stated scope. Section 3 computes the missing local ingredient: the local Whittaker factor of the N -scaled binary lattice at a split level prime, in closed form (Theorem 3.1), with the corollary that the empirical “level support rule” of the computational literature is exactly the vanishing of that factor (Corollary 3.2). Section 4 re-runs the case analysis with this factor and finds no new case (Proposition 4.1). Section 5 records the correction that the CM values require, computed

exactly, and Section 6 lists the explanations we have eliminated. Section 7 states what remains open.

All numerical statements below are exact rational computations, carried out in Magma; the supporting scripts are described in Appendix A.

2. THE CHAIN, AND THE SCOPE OF ITS CENTRAL STEP

Write $U = \lambda^\perp$ for the negative definite plane attached to a CM point of discriminant d , $L_+ = L \cap \mathbb{Q}\lambda$ and $L_- = L \cap U$. Following [6, (11),(12)], one sets

$$\kappa_\eta(m) = \sum_{\mu \in L/(L_+ + L_-)} \sum_{x \in \eta_+ + \mu_+ + L_+} \kappa_{\eta_- + \mu_-}^-(m - \langle x, x \rangle/2),$$

where $\kappa_\mu^-(m)$ is the s -linear coefficient of the m -th Fourier coefficient $A_\mu(s, m, v)$ of an incoherent Eisenstein series of weight 1 attached to L_- , and $\kappa_\mu^-(0) = 0$ unless $\mu = 0$.

For $m > 0$ one has, with $S_{m,\mu} = \{p : p \mid \Delta \text{ or } v_p(m) > 0\}$ and $\Delta = \text{disc}(L_-)$,

$$A_\mu(s, m, v) = -\frac{2\pi}{L(s+1, \chi_d)} \prod_{p \in S_{m,\mu}} \frac{W_{m,p}(s, \varphi_{\mu,p})}{1 - \chi_d(p)p^{-1-s}}, \quad (1)$$

a genuine Euler product because $\varphi_\mu = \text{char}(\mu + \widehat{L_-})$ is a pure tensor. Since $A_\mu(s, m, v)$ vanishes at $s = 0$, some factor must vanish there, and differentiating (1) yields [6, Lemma 16]: the derivative is the single term at the vanishing place, and vanishes outright if two or more factors vanish.

That step is correct as an identity. What it inherits from [3] is the identification of the vanishing of a local factor with a *representability* obstruction, i.e. with membership of $\text{Diff}(m)$. Kudla–Rapoport–Yang state the scope of their computation plainly:

“in section 2, we have considered only those Eisenstein series built from sections which are defined by the characteristic functions of the completions of $\mathcal{O}_k$. More general sections could be considered at the cost of (i) more elaborate calculations of Whittaker functions and their derivatives used in computing the Fourier expansion of the central derivative, and (ii) incorporation of a level structure in the moduli problem.” [3, §6]

(They also fix a maximal order $\mathcal{O}_B$.) The lattice L_- arising here is not of that shape. Write $\Delta = \text{disc}(L_-)$. For a discriminant d with $N \nmid d$ — the case in which the level prime is *unramified* in $k = \mathbb{Q}(\sqrt{d})$, and hence, by Eichler’s condition on optimal embeddings, necessarily *split* — one has $\text{ord}_N(\Delta) = 2$: the binary lattice is N -scaled at the level prime, not unimodular. This is exactly a “more general section”.

Remark 2.1. The distinction is invisible in the maximal-order setting: there L_- is unimodular at every split prime, no split prime can obstruct representability, and correspondingly the logarithms produced by [5, Theorem 4.1] run over ramified and inert places only. Every discriminant admitting a CM point on $X_0^D(N)$ with $N \nmid d$ has N split, so this configuration is precisely the one the maximal-order theory never meets.

3. THE LOCAL WHITTAKER FACTOR AT A SPLIT LEVEL PRIME

Part (i) of the programme quoted above, at the level prime, is the following computation. Let $X = N^{-s}$.

Theorem 3.1. *Let $d < 0$ be a discriminant with $N \nmid d$ and let L_- be the negative definite binary lattice attached to a CM point of discriminant d , so $\text{ord}_N(\text{disc } L_-) = 2$. Then for every $m \geq 1$, with $j = \text{ord}_N(m)$,*

$$W_{m,N}(X) = 1 + (N-1)(X + X^2 + \cdots + X^j) - X^{j+1}.$$

In particular

$$W_{m,N}(1) = (N-1) \text{ord}_N(m), \quad W'_{m,N}(1) = (j+1) \left(\frac{(N-1)j}{2} - 1 \right).$$

The statement is a computation rather than a proof: it has been verified in exact arithmetic for $N = 2, 3, 5$ and all $1 \leq m \leq 60$, on the lattices L_- of $X_0^{15}(2)$, $X_0^6(5)$ and $X_0^{10}(3)$ at firing discriminants, giving 180 independent checks. It is pinned as a regression test in the accompanying code.

Corollary 3.2. *$W_{m,N}(1) = 0$ if and only if $N \nmid m$.*

This identifies, and explains, an empirical rule that has been observed repeatedly in computations with these curves: the correction discussed in Section 5 is supported exactly on the indices m with $N \mid m$. That rule is the vanishing of the level-prime factor.

Remark 3.3. At $j = 0$ the factor is $W_{m,N}(X) = 1 - X = 1 - N^{-s}$, the inverse local zeta factor. Its vanishing at $s = 0$ is therefore of a different nature from a representability obstruction: it holds identically for all m with $N \nmid m$, and at a split N the local quadratic space represents every m .

Remark 3.4. The value $W_{m,N}(1) = (N-1) \text{ord}_N(m)$ coincides with the local factor Kudla–Yang obtain at $p \mid N$ for the *quaternary* Eichler lattice [4, (9.1)]. The agreement appears to be a genuine coincidence of shape between the quaternary lattice and the binary L_- ; we do not have an explanation for it.

4. RE-RUNNING THE CASE ANALYSIS

With Theorem 3.1 in hand one can redo the enumeration of [3, Prop. 2.8] for the N -scaled section. Normalising as in (1), put $f_p(s) = W_{m,p}(s)/(1 - \chi_d(p)p^{-1-s})$.

Proposition 4.1. *At a firing discriminant, f_N has a simple zero at $s = 0$ exactly when $N \nmid m$, and this zero is not a Diff zero. Consequently*

$$\text{ord}_{s=0} A_\mu(s, m, v) = |\text{Diff}(m)| + [N \nmid m],$$

and since incoherence forces $|\text{Diff}(m)| \geq 1$:

- (1) *if $N \mid m$, the order is $|\text{Diff}(m)|$, and the derivative is nonzero precisely when $|\text{Diff}(m)| = 1$, contributing a logarithm at the ramified or inert place of $\text{Diff}(m)$;*
- (2) *if $N \nmid m$, the order is at least 2 and the derivative vanishes.*

In neither case does a log N occur.

So the level structure only ever raises the order of vanishing: it removes contributions, and creates no new contributing configuration. The three cases of [3, Prop. 2.8] stand, with no fourth. This is confirmed computationally: over roughly 190 terms (μ, x) arising in the evaluation of $\kappa_0(m)$ on three bases, the level prime is *never* the unique vanishing place, while it lies among two or more in 60–80% of terms.

Corollary 4.2. *Within this framework, a log N can enter a CM value only through the $m = 0$ term $c_0(0)\kappa_0(0)$, where $\kappa_0(0)$ carries the summand $\sum_{p \mid N/(N,d)} \log p$ of [6, Remark 15(2), Lemma 20].*

5. THE CORRECTION THE CM VALUES REQUIRE

Against this stands a computation of the input form itself. Following [2, eq. (6)] one may form $F_f = \sum_{\gamma \in \Gamma_0(M) \backslash SL_2(\mathbb{Z})} (f|_{1/2}\gamma) \rho(\gamma^{-1}) e_0$ and evaluate it numerically; the metaplectic phase is handled by evaluating both the slash and ρ along one and the same ST -word, so that they share a single metaplectic lift. Every such evaluation is gated on reproducing the principal part of F_f predicted by [2, Lemma 24] from the two scalar q -expansions of f ; on the runs reported here that gate is met to 10^{-103} .

Observation 5.1. For every Borchers form produced by the pipeline on $X_0^{15}(2)$, $X_0^6(5)$, $X_0^{10}(3)$ and $X_0^{21}(2)$ one finds $c_0(0) = 0$ (to 10^{-103}), so the hypothesis $c_0(0) = 0$ of [2, Thm. B] is satisfied and, by Corollary 4.2, no $\log N$ can arise.

Observation 5.2. The isotropic cosets of $L^\vee/L$ number exactly $2N - 1$; the D -part of the discriminant form is anisotropic, so all of them come from the level- N hyperbolic plane. All $2N - 2$ nonzero isotropic cosets carry the same constant term $c_\eta(0)$.

Observation 5.3. Let $\text{mult}(f) = \frac{1}{2}c_\eta(0)$ for any nonzero isotropic η . Then adding $\text{mult}(f) \cdot \sum_{p|N/(N, d_{\text{fund}})} \log p$ to the computed CM value reproduces, exactly, the corrections independently measured by kernel-consistency on $X_0^{15}(2)$ (9 values), $X_0^6(5)$ (5) and $X_0^{10}(3)$ (5): 19 of 19. A fourth base $X_0^{21}(2)$ supplies nine further values, with no prior ground truth against which to check them.

That the correction is genuinely required, and not an artefact of a normalisation, is visible on $X_0^6(19)$: on a set of CM points including discriminants divisible by N , the uncorrected values are mutually *inconsistent* — the associated system has no solution — whereas with the correction the published model is recovered.

Writing $\text{mult}(f) = \sum_m c(-m)A_m$ and using Borchers' obstruction to set $A_m = -b(m)/4$ with b the relevant weight $3/2$ Eisenstein coefficient, the linear functional is uniquely determined per base by the measured multipliers, with 4–6 spare conditions in each case. Table 1 records it.

base	A_m (equivalently $b(m) = -4A_m$)		
$X_0^{15}(2)$	$A_2 = -1$	$A_{10} = 1$	$A_{30} = 0$
$X_0^6(5)$	$A_{10} = 3/2$	$A_{15} = 1/2$	$A_{30} = 3/2$
$X_0^{10}(3)$	$A_3 = 1/2$	$A_{12} = 1/2$	$A_{30} = 3/2$
$X_0^{21}(2)$	$A_2 = -2, A_6 = -4$	$A_{18} = 2, A_{42} = -8$	(0-block: $B_{21} = -5$)

TABLE 1. The correction, per index, as a linear functional of the principal part. The values are d -independent: at every firing discriminant the coefficient of $\log N$ in $\kappa_0(m)$ must equal A_m , whereas the recipe of §2 returns 0.

6. EXPLANATIONS ELIMINATED

Each of the following was tested against the data of Section 5 and is inconsistent with it. We record them because several are natural first guesses.

- (1) $\text{mult} = c_0(0)$, or any linear combination of the two scalar constant terms $\text{Coeff}(f_\infty, 0)$ and $\text{Coeff}(f_0, 0)$: inconsistent on three bases.
- (2) Evaluating the local Whittaker at a nonzero isotropic coset η^* rather than at 0: destroys the pattern of vanishing indices on all three bases.

- (3) Replacing the local factor at $p \mid D$ by the anisotropic branch of [4, Prop. 5.3]: does not reproduce b , under either sign of κm . (The implementation of [4, Props. 5.3, 5.4] agrees with the pipeline’s own factor at every good prime, 470 checks, and at the level prime, 150 checks.)
- (4) The incoherent support condition $|\text{Diff}(m)| = 1$: scores 8/9 on the exact coefficients and is the only rule that predicts $b(30) = 0$ on $X_0^{15}(2)$, but is far too permissive — for $r \leq 60$ it holds at 15, 19 and 20 indices with $N \nmid r$, where the coefficient must vanish.
- (5) Cancelling the zeta factor $1 - X$ at the level prime before counting vanishing places: the recovered terms differentiate at ramified and inert primes, never at N .
- (6) The inner $m = 0$ of κ_η , i.e. terms with $\langle x, x \rangle / 2 = m$: never occurs, in all 60 (discriminant, index) pairs over every firing discriminant of the three bases.
- (7) Schofer’s outer $m = 0$ term: vanishes by Observation 5.1.
- (8) The re-enumeration of §4: adds no case.

7. THE OPEN QUESTION

Corollary 4.2 and Observation 5.1 are in direct tension with Observation 5.3 and with the failure of the uncorrected computation on $X_0^6(19)$. We see two possibilities and cannot presently distinguish them.

- (1) The constant term $c_0(0)$ of the form F actually attached to ψ_F differs from that of the coset sum F_f . The principal parts agree to 10^{-103} , but the principal part alone does not determine $c_0(0)$.
- (2) The correction compensates an error elsewhere in the computation of the CM values, outside Schofer’s formula. It is a d -independent linear functional of the principal part, which is the shape of a per-form normalisation.

In either case Theorem 3.1 and Proposition 4.1 stand, and part (ii) of the Kudla–Rapoport–Yang programme — the incorporation of a level structure — remains to be carried out. Table 1 is what any corrected formula must reproduce, index by index.

APPENDIX A. REPRODUCIBILITY

The computations were carried out in Magma against the codebase accompanying this note.

- **VectorValuedForm.m** constructs F_f : coset representatives for $\Gamma_0(M) \backslash SL_2(\mathbb{Z})$, ST -words, the weight 1/2 metaplectic slash, the Weil representation over $\mathbb{C}$, and the Fourier extraction. The action of $\rho(S)$ is implemented as a finite Fourier transform on $L^\vee/L$, which reduces the cost from $|G|^2$ to $|G| \sum_r d_r$ and the memory from $|G|^2$ to $|G|$; without this the case $|G| = 24200$ is out of reach.
- **EisensteinLocalFactors.m** implements [4, (2.16), (2.17), Props. 5.3, 5.4] and exposes **LocalWhittakerPolynomial**, the Whittaker polynomial whose s -derivative produces the logarithms.
- Numerical parameters matter. The horocycle height must satisfy $\Im \tau \geq 1$, or some coset is carried into the pole at ∞ and the terms overwhelm their sum. The number of sample points must beat the positive Fourier coefficients, which grow like $\exp(4\pi\sqrt{pm})$ for pole order p : on the pole-order-30 forms of $X_0^{15}(2)$ the principal-part gate reads $5.8 \cdot 10^{73}$, $1.2 \cdot 10^{62}$, $2.7 \cdot 10^{28}$ and $8.7 \cdot 10^{-103}$ at $K = 48, 64, 96$ and 192 respectively. Raising the working precision without raising K has no effect.
- Every reported value is gated on the principal-part check; a run failing that gate is discarded.

REFERENCES

- [1] E. Errthum, *Singular moduli of Shimura curves*, Canad. J. Math. **63** (2011), 826–861.
- [2] J.-W. Guo and Y. Yang, *Equations of hyperelliptic Shimura curves*, Proc. Lond. Math. Soc. (2017), doi:10.1112/S0010437X16007739.
- [3] S. Kudla, M. Rapoport and T. Yang, *On the derivative of an Eisenstein series of weight one*, Internat. Math. Res. Notices (1999), no. 7, 347–385.
- [4] S. Kudla and T. Yang, *Eisenstein series for $SL(2)$* , Sci. China Math. **53** (2010), 2275–2316.
- [5] J. Schofer, *Borcherds forms and generalizations of singular moduli*, J. reine angew. Math. **629** (2009), 1–36.
- [6] Y. Yang, *Special values of hypergeometric functions and periods of CM elliptic curves*, arXiv:1503.07971.